Snap-Conditioned Retrieval for Legal RAG: A Four-Benchmark Bottleneck Study

Hamza Iqbal
Washington University in St. Louis
hiqbal@wustl.edu

Hanson Li
Washington University in St. Louis
hanson.l@wustl.edu

Josh Li
Washington University in St. Louis
josh.l@wustl.edu

Abstract

Legal retrieval-augmented generation is usually summarized as one accuracy number, but legal QA failures are not one phenomenon. We study snap-conditioned HyDE across BarExamQA, HousingQA, LegalBench-SCALR, and CaseHOLD, using one full-corpus legal result and paired $N = 200$ diagnostics across four legal benchmarks to probe RAG bottlenecks. On full-corpus BarExamQA, `rag_snap_hyde` improves Gemma 4 26B from 78.08% to 81.17% (+3.09pp, McNemar $p = 0.0130$) and Gemma 4 E4B from 58.49% to 62.18% (+3.68pp, $p = 0.0129$). The rest of the evidence is deliberately mixed: BarExamQA is retrieval-depth flat, SCALR is candidate-depth sensitive then saturates at top-5, HousingQA improves with metadata-filtered retrieval, and CaseHOLD improves gold-option retrieval without a reliable answer lift. The contribution is therefore not a universal new RAG method, but a bottleneck-typed analysis showing when legal RAG needs more candidates, better query generation, answer anchoring, metadata filtering, or stronger evidence-to-option conversion.

1 Introduction

Legal QA is a natural target for retrieval augmentation. Many questions depend on statutes, holdings, jurisdiction, procedural posture, and precise rule exceptions that may not be reliably stored in a model's parameters. However, legal RAG is also a setting where more retrieved text can easily hurt. A passage can be topically related but legally incomplete; a holding can mention the same doctrine while being the wrong option; a state statute can be right in subject matter but wrong in jurisdiction. These errors are not just retrieval failures or generation failures. They are interactions between retrieval, reasoning, and answer selection.

Our initial project pursued a broader agentic legal RAG system: route the question, decompose the issue, retrieve per gap, judge sufficiency, and replan. That design matches legal research practice, but it is difficult to evaluate scientifically because any gain could come from retrieval, extra tokens, a second reasoning pass, or prompt luck. We therefore moved from full-system complexity to atomic retrieval interventions.

The central method family is snap-conditioned HyDE. A "snap" is a first-pass answer or reasoning trace. A HyDE-style hypothetical passage is then generated from that snap and used for retrieval. In the core sequential and two-call HyDE modes, the final answer sees the question and retrieved evidence; the snap shapes retrieval rather than being handed directly to final synthesis. Separate snap-only and one-call ablations test what happens when visible snap reasoning is itself part of the final answer path. The intuition is legal: before asking the model to read external authority, ask it to state the issue and likely rule. Retrieval is then shaped by that legal frame rather than by the raw question alone.

The current legal-only evidence is not uniformly positive, and that is useful. BarExamQA shows a real full-corpus snap-HyDE lift, but top-1 versus top-5 retrieval is flat, so the lift should not be described as "more documents help." LegalBench-SCALR shows a strong top-1 collapse and top-5 saturation, which suggests candidate-set size matters more than pseudo-document query formulation. HousingQA has a promising top-10 signal and, after fixing a metadata-casing bug, state-filtered retrieval reaches 61.5% at k=5, beating generic top-5 retrieval and directionally beating generic top-10. CaseHOLD now has repaired gold-option retrieval, and snap-HyDE raises gold retrieval from 16.0% to 47.0%, but answer accuracy only moves from 69.5% to 72.0% with $p = 0.4421$.

We exclude MuSiQue from the main report because it is not a legal benchmark. The final class report is framed around four legal tasks; non-legal multi-hop QA remains only an internal mechanism check.

2 Related Work

Retrieval-augmented generation combines parametric generation with retrieved non-parametric evidence [Lewis et al., 2020]. HyDE generates a hypothetical document and embeds it for zero-shot retrieval [Gao et al., 2023], while Query2Doc expands queries with LLM-generated pseudo-documents [Wang et al., 2023]. FLARE retrieves during generation by using predicted future content as a query [Jiang et al., 2023]. Our method family inherits directly from this line: generated text is used as a retrieval handle, not as trusted evidence.

Adaptive RAG work asks when or how retrieval should be invoked. Self-RAG learns to retrieve and critique through reflection tokens [Asai et al., 2024]. Corrective RAG grades retrieved evidence and triggers correction when retrieval is unreliable [Yan et al., 2024]. Adaptive-RAG routes questions by complexity [Jeong et al., 2024]. Speculative RAG drafts multiple answers from retrieved subsets and verifies them with a larger model [Wang et al., 2025]. These systems motivate our future direction, but we do not claim to implement them here. Instead, we use their shared lesson: retrieval should be conditional on observed failure mode.

Legal NLP benchmarks increasingly show that legal RAG cannot be reduced to generic open-domain QA. LegalBench provides a broad suite of legal reasoning tasks, including SCALR holding selection [Guha et al., 2023]. CaseHOLD evaluates whether a model can select the holding referenced by a court-opinion context [Zheng et al., 2021]. Zheng et al. introduce BarExamQA and Housing Statute QA to evaluate realistic legal retrieval and answering workflows [Zheng et al., 2025]. LegalBench-RAG focuses on retrieving minimal relevant legal snippets [Pipitone and Alami, 2024], and Legal RAG Bench provides an end-to-end legal RAG benchmark with factorial retriever/generator decomposition [Butler and Butler, 2026]. RAGChecker similarly argues for fine-grained separation of retrieval and generation errors [Ru et al., 2024]. Our work follows this diagnostic turn: the same RAG method should be judged differently depending on whether the active failure is retrieval depth, candidate-set selection, metadata filtering, or evidence conversion.

We also considered hypothetical-representation work beyond HyDE. HyQE ranks contexts with hypothetical query embeddings [Zhou et al., 2024]. We did not implement a separate "HyRE" or hypothetical reasoning embedding index in this report. The implemented object is simpler: online snap-conditioned HyDE used as a legal retrieval probe.

3 Framework Design

3.1 Method ladder

The report uses an inheritance-style method ladder. Each method adds one capability where possible, so negative rows remain interpretable.

Method	What it isolates
`llm_only`	Parametric legal answer without retrieval.
`snap_only_in_final`	Whether a first-pass answer/reasoning trace helps without retrieval.
`rag_simple`	Standard retrieval plus answer generation.
`rag_hyde`	Question-conditioned hypothetical document retrieval.
`rag_snap_hyde`	Snap-conditioned HyDE retrieval plus final synthesis.
`rag_snap_hyde_1call`	Whether snap reasoning alone after ordinary retrieval is enough.
`rag_snap_hyde_2call`	Fixed-cost snap and HyDE passage in one first call, final synthesis in a second.
`multi_hyde_diverse`	Multiple generated retrieval passages for query diversity.
`subagent_rag`	Gap-routed report generation; currently used as a negative over-abstention control.

3.2 Snap-conditioned HyDE

Figure 1 shows the core method. The model first writes a snap answer or legal analysis. It then writes a hypothetical passage that states the likely controlling rule, fact, or holding. That passage is embedded and used for retrieval. The final model sees retrieved passages along with the original question; the snap is mediated through the retrieval query unless the run is an explicit snap-visible ablation.

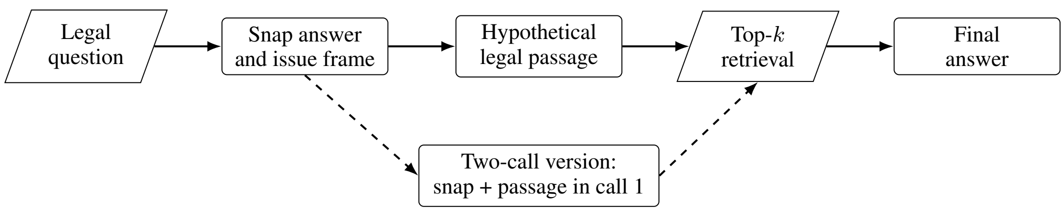

Figure 1: Snap-conditioned HyDE. The generated passage shapes retrieval; the retrieved corpus passages remain the external evidence used by final synthesis.

3.3 Bottleneck diagnostics

We use three families of diagnostics. First, top-k sensitivity tests whether answer accuracy depends on retrieving more than the single best passage. This is a retrieval-policy stress test, not pure context truncation, because the cross-encoder reranks a first-stage candidate pool before selecting the final top-k. Second, snap-only and HyDE-only ablations ask whether the gain comes from answer anchoring, generated-query retrieval, or their combination. Third, gold-retrieval and disagreement audits ask whether better retrieval actually converts into better final answers.

The four legal benchmarks map naturally onto different bottleneck hypotheses: BarExamQA tests answer anchoring under strong legal priors, HousingQA tests state-specific legal retrieval, LegalBench-SCALR tests small candidate-set holding selection, and CaseHOLD tests whether a retrieved holding can be converted into the correct answer option.

4 Experimental Setup

Benchmarks. BarExamQA contains 1,195 bar-exam multiple-choice questions with fact patterns and legal passages. HousingQA contains yes/no housing-law questions over state statutes. LegalBench-SCALR contains 571 Supreme Court holding-selection questions. CaseHOLD contains court-opinion contexts and five candidate holdings. We deliberately exclude non-legal MuSiQue from the main benchmark set, although it remains useful for internal multi-hop stress testing.

Models. The full-corpus BarExamQA rows use Gemma 4 26B-A4B and Gemma 4 E4B served through the shared evaluation harness. LegalBench-SCALR and CaseHOLD diagnostic rows use Llama 70B. HousingQA uses Gemma 4 26B. For paired claims, we compare runs on the same question slice and provider. Claims are scoped to these model/provider settings rather than presented as model-independent benchmark rankings.

Metrics. We report accuracy for multiple-choice and yes/no datasets. Paired deltas use exact McNemar tests with b/c discordance counts. Retrieval diagnostics report gold retrieval only when the dataset's gold mapping and corpus collection are known to be valid. The table rows and paired deltas in this draft were re-checked against detail logs in `reports/final_class_report/evidence_snapshot.md`.

5 Results and Analysis

5.1 BarExamQA: full-corpus snap-HyDE lift

BarExamQA is the strongest legal positive result. On Gemma 4 26B-A4B, `rag_snap_hyde` reaches 81.17%, compared with 78.08% for `rag_simple` (+3.09pp, b/c=124/87, $p = 0.0130$). On Gemma 4 E4B, the same method improves 58.49% to 62.18% (+3.68pp, b/c=172/128, $p = 0.0129$).

Table 1: BarExamQA full-corpus Gemma 4 26B-A4B method matrix, $N = 1195$.

Method	Accuracy	Δ vs `rag_simple`
`rag_simple`	78.08%	–
`golden_passage`	78.66%	+0.58pp
`rag_hyde`	78.91%	+0.83pp
`llm_only`	79.75%	+1.67pp
`snap_only_in_final`	80.59%	+2.51pp
`rag_snap_hyde`	**81.17%**	**+3.09pp**

The mechanism is not pure retrieval depth. On a paired $N = 200$ diagnostic, `rag_simple` top-5 is 82.5% and top-1 is 83.0% ($p = 1.0$). The full-corpus lift is better read as snap-conditioned query anchoring plus evidence reconciliation, with model-size dependence: the 26B audit suggests the first-pass legal frame is already strong, while retrieval/synthesis appears more useful on the smaller E4B model. A legal MC model can have strong prior knowledge of the rule; retrieval then helps only if it is framed in a way that does not overwrite the right legal analysis with a partial snippet. Figure 2 shows this cross-size pattern.

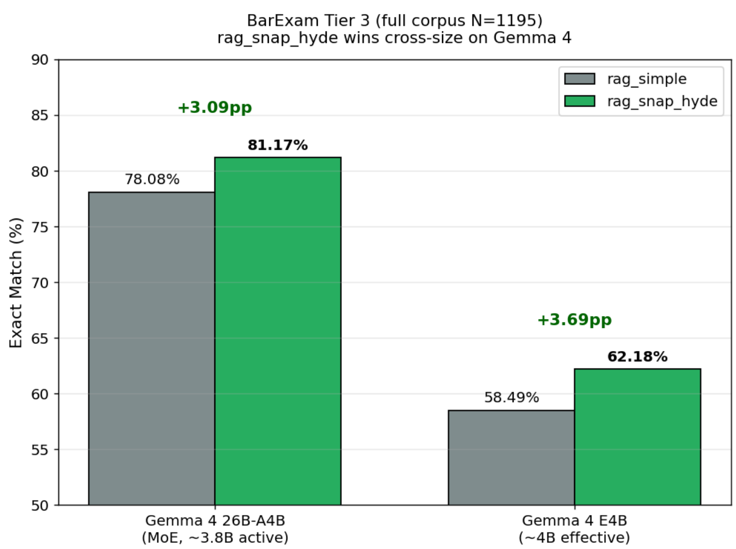

Figure 2: BarExamQA cross-size result. `rag_snap_hyde` improves over `rag_simple` on both Gemma 4 model sizes. The snap conditions retrieval; visible snap reasoning is tested by separate controls.

5.2 Four-benchmark legal bottleneck matrix

Table 2 is the compact result summary. It intentionally includes negative and directional rows. The current evidence does not show one method winning everywhere; it shows which legal bottleneck each dataset is probing.

Table 2: Legal-only benchmark matrix. MuSiQue is excluded because it is not legal-domain data.

Benchmark	Active bottleneck	Best current evidence	Caveat
BarExamQA / Gemma 4	Snap-conditioned query anchoring under strong legal priors	Full $N = 1195$: snap-HyDE is +3.09pp at 26B and +3.68pp at E4B; top-5 82.5% vs top-1 83.0% on $N = 200$	Not a pure retrieval-depth win.
HousingQA / Gemma 4	Depth, answer bias, and jurisdiction filtering	Top-1 50.5% to top-10 58.0% (+7.5pp, $p = 0.0722$); k=5 state filter is 61.5%	State filter beats generic top-5 ($p = 0.044$); k=10 state filter is still chunking.
LegalBench-SCALR / Llama 70B	Candidate-set size	Top-1 59.5%, top-5 77.0%, top-10 77.0%; HyDE 77.5%; diverse HyDE 75.5%	Depth helps to top-5; generated queries do not yet add answer lift.
CaseHOLD / Llama 70B	Evidence-to-option conversion	Top-1 64.5%, top-5 69.5%, top-10 68.0%; HyDE and two-call both 72.0%	Better retrieval recall is not yet a reliable answer lift.

Figure 3 summarizes the key diagnostic split: the same top-k or HyDE intervention has different meaning depending on whether accuracy and gold retrieval move together.

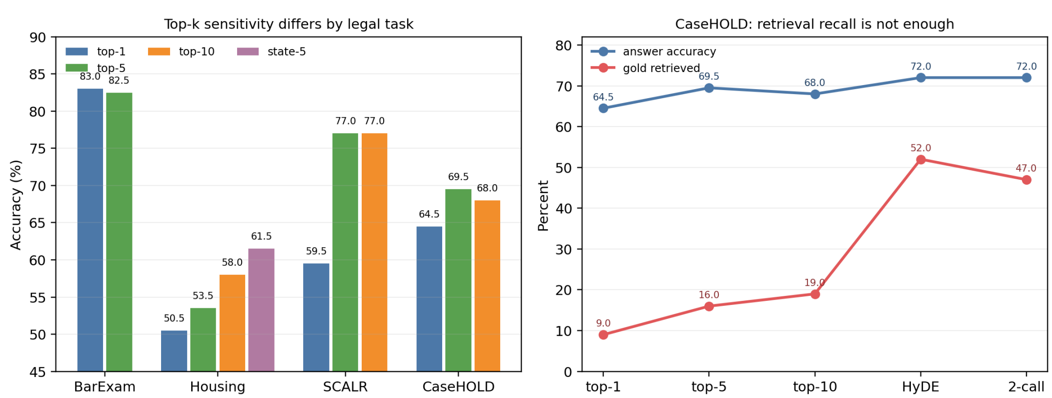

Figure 3: Legal benchmarks expose different bottlenecks. BarExamQA is depth-flat, SCALR benefits sharply from top-5 candidates, HousingQA improves with state-filtered retrieval, and CaseHOLD shows that gold-option retrieval can improve faster than answer accuracy.

The most important negative result is CaseHOLD. After repairing the holdings corpus and gold mapping, top-1 `rag_simple` reaches 64.5% with 9.0% gold retrieval, while top-5 reaches 69.5% with 16.0% gold retrieval (+5.0pp, $p = 0.0525$). Top-10 retrieves more gold options (19.0%) but drops to 68.0%. HyDE retrieves the gold option in 52.0% of rows and two-call snap-HyDE in 47.0%, but both answer at 72.0%; neither is a significant lift over top-5 `rag_simple`. If retrieval recall were sufficient, answer accuracy should improve sharply. It does not. This suggests an evidence-to-option bottleneck: retrieved holdings can still be paraphrased distractors, overly generic, or hard to map back to the exact candidate label.

SCALR gives a different holding-selection story. Top-1 is poor, top-5 fixes most of the loss, top-10 adds no answer accuracy beyond top-5, and a validated `rag_hyde` follow-up is also flat at 77.5% despite retrieving the gold holding in 66.0% of rows. A one-call snap-final variant drops to 70.5% ($p = 0.00235$ versus top-5), so simply adding a snap rationale to the final answer is not a free improvement on this task. Diverse HyDE also stays below top-5 at 75.5%. This is not a generic "legal holdings ignore retrieval" result. It says SCALR needs a small candidate set, while CaseHOLD needs better evidence conversion.

HousingQA is the strongest evidence that legal metadata is not optional. The first state-filter experiment failed because question states were display-case strings while Chroma metadata was lowercase; that failed row is rejected as empty retrieval. After repairing the casing bug, k=5 state-filtered retrieval lands at 61.5%, with zero empty retrieval rows and 81/200 gold statutes retrieved. This is +8.0pp over generic top-5 `rag_simple` (b/c=36/20, $p = 0.044$) and +3.5pp over generic top-10 ($p = 0.435$). The top-10 state-filter leg is still running in deterministic chunks, so the current claim is not that state filtering is fully optimized; it is that the previous "Housing is blocked" story has become a concrete metadata-filtering signal.

5.3 BarExam single gold passage is a noisy control

The BarExamQA golden-passage control appears paradoxical: providing one gold passage reaches only 78.66%, below `llm_only` at 79.75% and `snap_only_in_final` at 80.59%. The reason is legal evidence granularity. A labeled passage can be topically correct but not decisive for the fact pattern; it can also highlight a distractor doctrine. Snap-first reasoning gives the model an initial legal frame before evidence reconciliation, which partly protects against over-reading one snippet. This claim is specific to the BarExamQA single-passage control; SCALR behaves differently, with high conditional accuracy when the gold holding is retrieved. Figure 4 visualizes the paired flips behind this control.

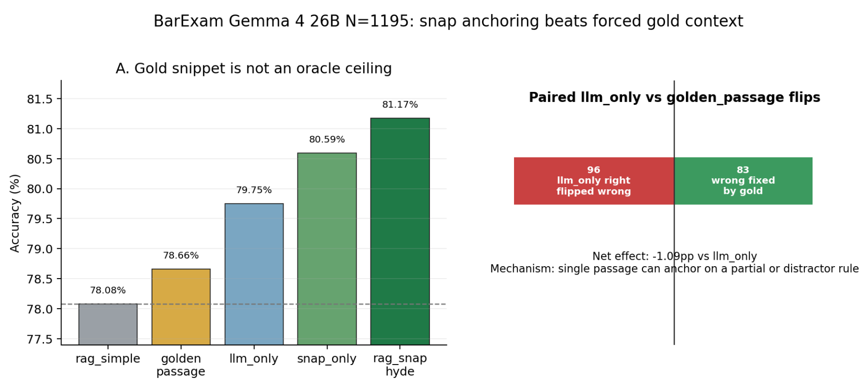

Figure 4: BarExamQA mechanism audit. A single gold passage is a noisy control rather than an oracle ceiling; snap-first reasoning can recover some cases where forced context distracts the model.

5.4 Cost and budget behavior

The ablations also expose when extra calls and tokens are worth spending. Figure 5 plots paired $N = 200$ legal slices with marker size proportional to total tokens per question. BarExamQA is the cleanest case for snap-style reasoning budget: the two-call version reaches 85.5%, nearly matching the sequential snap-HyDE row at 86.0%, while using fewer calls and fewer tokens per question. HousingQA behaves differently: generic top-10 retrieval is one call but expensive in total tokens (4.6k tokens per question), while two-call snap-HyDE is slightly lower accuracy at a smaller 3.2k tokens per question. State-filtered k=5 is now the stronger Housing candidate because it uses jurisdiction metadata directly rather than spending generic depth. The top-10 row also reduces a top-1 tendency to over-predict "Yes" on a dataset whose gold labels are mostly "No," so the generic depth gain should not be over-described as jurisdiction repair. SCALR and CaseHOLD show the main caution: two-call reasoning roughly doubles calls and tokens, but SCALR does not improve and CaseHOLD improves only modestly. This is why we frame method choice as budgeted routing, not as always-on snap-HyDE.

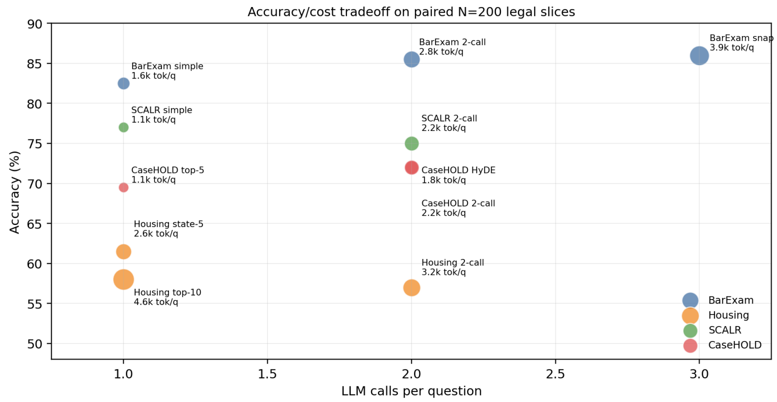

Figure 5: Calls and tokens per question matter. Extra reasoning budget is useful on BarExamQA, ambiguous on HousingQA and CaseHOLD, and not currently justified on SCALR. Token values are computed from the paired detail logs.

5.5 Open validations

Several validations are still in flight and should not be promoted as final numbers until their detail logs pass the same checks. HousingQA now has a clean k=5 state-filter row, but the k=10 state-filter leg is still chunking; it will decide whether metadata filtering simply recovers the generic top-10 gain or continues improving beyond it. SCALR follow-ups have now landed locally as flat or negative, so the next SCALR question is not more query generation but option grounding or reranking.

6 Discussion: From Method Search to Bottleneck Routing

The most useful interpretation of these results is not that snap-HyDE is the final method. It is that a small family of fixed interventions can expose which part of a legal RAG pipeline is currently binding. This matters because many RAG papers collapse several different phenomena into one comparison: retrieval versus no retrieval. In our logs, that comparison is too coarse. LegalBench-SCALR loses heavily at top-1 and recovers at top-5, so a route that expands the candidate set is plausible. BarExamQA is top-k flat but improves with snap-conditioned HyDE, so the route should preserve an answer frame rather than simply retrieve more passages. HousingQA likely requires metadata-constrained retrieval checks, because the question state is part of the legal problem and a normalized state filter now beats generic top-5 retrieval. CaseHOLD improves gold retrieval under two-call snap-HyDE but not answer accuracy, so the route should focus on mapping retrieved holdings back to answer options.

This gives a feasible version of the "agentic" idea without making the agent itself the unit of evaluation. Instead of asking a planner to invent an open-ended research workflow, we can ask a controller to choose among a small set of evidence-budgeted interventions. The controller would observe cheap features before or during retrieval: question type, answer format, available metadata, top-score margin, retrieved-document agreement, candidate-option similarity, and whether the snap answer agrees with retrieval. It would then choose a route: ordinary RAG, top-k expansion, snap-conditioned HyDE, state-filtered retrieval, multi-query diversity, or option-aware reranking. The result is still agentic in the sense that method adaptation happens on the fly, but it is experimentally cleaner because every route corresponds to a measured ablation cell.

Observed signal	Likely bottleneck	Route to test next
Top-1 fails, top-5 recovers, top-10 flat	Candidate-set size rather than answer reasoning	Use top-5 as default; spend budget on reranking or option grounding instead of deeper context.
Snap-only beats or matches retrieval	Evidence is distracting or under-specified relative to legal priors	Keep a first-pass legal frame in final synthesis and audit whether evidence overwrites correct priors.
Generated retrieval improves gold-hit but not answer accuracy	Evidence-to-answer conversion bottleneck	Add answer-option-aware synthesis, contradiction checks, or verifier scoring rather than more retrieval.
Question contains jurisdiction metadata	Corpus filtering and metadata normalization bottleneck	Filter before retrieval, validate non-empty evidence, and compare against generic top-k retrieval.
Retriever scores are low or retrieved passages disagree	Query formulation bottleneck	Try snap-HyDE or diverse HyDE passages, then measure whether gold retrieval and answer accuracy move together.

This framing also explains why the current negative results are useful. A negative CaseHOLD answer delta after a positive gold-hit delta is not just a failed method row; it identifies a downstream conversion problem. A flat BarExamQA top-k sweep does not make retrieval irrelevant; it says that retrieval depth is not the active lever for that slice. The HousingQA state-filter repair shows why metadata gates belong in the scientific evaluation path: a casing mismatch produced plausible-looking parametric answers with zero retrieved evidence, while the repaired filter changed the method result. These are the hooks for a stronger final project: tune snap-HyDE quality where query formulation is binding, tune metadata filters where jurisdiction is binding, and tune verifier or option-mapping logic where evidence conversion is binding.

The class-report claim is therefore deliberately modest but concrete. We built and audited a legal RAG harness that can run comparable methods across several legal benchmarks; we found one full-corpus legal snap-HyDE lift on BarExamQA; we found several $N = 200$ diagnostic signals that disagree with a single-method story; and we turned those signals into a routing agenda. That is a better foundation for future work than another broad sweep over loosely comparable prompts, because it says what should change when a method fails.

7 Limitations and Future Work

The report is intentionally conservative. Only BarExamQA has a full-corpus legal result; HousingQA, SCALR, and CaseHOLD are paired $N = 200$ diagnostic slices for mechanism discovery rather than final benchmark claims. HousingQA has a repaired k=5 state-filter result, but its k=10 state-filter leg is still running and should be audited before being used as a table row.

The method set is incomplete relative to the literature. We define Self-RAG, CRAG, and Speculative RAG, but do not reimplement them or log true draft/verifier metrics. The next experiments should stay legal-only: finish HousingQA k=10 state-filtered retrieval, test SCALR option grounding or reranking, test CaseHOLD option-aware synthesis after the gold-retrieval lift, and wire LegalBench-RAG or Legal RAG Bench rather than adding another generic multi-hop dataset.

8 Conclusion

The legal-only evidence supports a bottleneck-typed view of RAG. Snap-HyDE helps BarExamQA at full corpus, but BarExamQA is not retrieval-depth limited. SCALR is candidate-depth limited, HousingQA shows a real metadata-filtering signal, and CaseHOLD shows that better generated-query retrieval does not automatically convert to better answers. These mixed results are not a failure of the project. They are the project signal: before building a bigger agent or declaring a RAG method better, we need to know whether the legal task needs more candidates, better generated queries, state-aware filtering, answer anchoring, or a stronger evidence-to-option mapper.

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